

Healthcare Readmission Analysis Report

Project Overview

Hospital readmissions are a major concern for healthcare providers as they impact patient outcomes, hospital performance, and healthcare costs. The objective of this project was to analyze patient readmission patterns and identify factors associated with 30-day hospital readmissions. The findings can support hospitals in improving discharge planning and reducing avoidable readmissions.

Dataset Summary

The dataset contains over 100,000 diabetic patient encounters collected from multiple hospitals in the United States over a ten-year period. Each record represents a hospital encounter and includes information related to patient demographics, admission details, hospital utilization, diagnoses, medications, and readmission status.

Key Variables

- Patient Demographics: Race, Gender, Age
- Admission Information: Admission Type, Admission Source, Discharge Disposition
- Hospital Stay Metrics: Time in Hospital, Number of Lab Procedures, Number of Procedures
- Utilization History: Number of Outpatient, Emergency, and Inpatient Visits
- Clinical Information: Number of Diagnoses
- Target Variable: Readmission Status (<30 Days, >30 Days, No Readmission)

Data Cleaning and Preparation

The dataset was cleaned and transformed to improve data quality and support analysis.

Data Cleaning Steps

- Replaced missing value indicators ("?", "N/A") with null values.
- Removed columns with excessive missing values.
- Removed unnecessary identifiers and medication-specific attributes.
- Handled invalid records and duplicates.
- Standardized column names and data types.

Feature Engineering

The following features were created:

- Readmission 30 Days Flag
- Total Prior Visits
- Long Stay Flag
- Polypharmacy Flag
- High Utilizer Flag

Business Questions

1. What is the overall distribution of readmission outcomes?

	readmitted	patient_count	percentage
1	NO	54861	53.9100000000000
2	>30	35545	34.9300000000000
3	<30	11357	11.1600000000000

2. Which age groups have the highest 30-day readmission rates?

	age	total_patients	readmitted_patients	readmission_rate
1	[20-30]	1657	236	14.24
2	[80-90]	17197	2078	12.08
3	[70-80]	26066	3069	11.77
4	[30-40]	3775	424	11.23
5	[60-70]	22482	2502	11.13
6	[90-100]	2793	310	11.1
7	[40-50]	9685	1027	10.6
8	[50-60]	17256	1668	9.67
9	[10-20]	691	40	5.79
10	[0-10]	161	3	1.86

3. Does a longer hospital stay increase readmission risk?

	time_in_hospital	total_patients	readmission_rate
1	1	14206	8.18
2	2	17224	9.94
3	3	17756	10.67
4	4	13924	11.81
5	5	9966	12.03
6	6	7539	12.59
7	7	5859	12.83
8	8	4390	14.24
9	9	3002	13.72
10	10	2342	14.35
11	11	1855	10.51
12	12	1448	13.33
13	13	1210	12.31
14	14	1042	12.96

4. How do prior healthcare visits relate to readmission?

	readmitted	avg_inpatient_visits	avg_outpatient_visits	avg_emergency_visits	avg_total_prior_visits
1	NO	0	0	0	0
2	>30	0	0	0	1
3	<30	1	0	0	2

5. Which admission types have the highest readmission rates?

	admission_type_id	total_patients	readmission_rate
1	1	53988	11.52
2	2	18480	11.18
3	6	5291	11.08
4	3	18868	10.39
5	5	4785	10.34
6	4	10	10
7	8	320	8.44
8	7	21	0

6. Does discharge disposition affect readmission likelihood?

	discharge_disposition_id	total_patients	readmission_rate
1	12	3	66.67
2	15	63	44.44
3	9	21	42.86
4	28	139	36.69
5	22	1992	27.71
6	5	1184	20.86
7	2	2128	16.07
8	3	13954	14.66
9	24	48	14.58
10	7	623	14.45
11	8	108	13.89
12	4	815	12.76
13	6	12902	12.7
14	18	3691	12.44
15	25	989	9.3
16	1	60232	9.3
17	23	412	7.28
18	14	372	6.45
19	13	399	4.76
20	16	11	0
21	10	6	0
22	19	8	0
23	27	5	0
24	17	14	0
25	11	1642	0

7. Is medication burden associated with higher readmission risk?

	num_medications	total_patients	readmission_rate
1	1	262	4.2
2	2	470	8.3
3	3	900	7.22
4	4	1417	8.05
5	5	2017	7.44
6	6	2698	8.49
7	7	3484	8.75
8	8	4353	9.56
9	9	4913	9.81
10	10	5346	9.97

8. Do patients with more diagnoses experience higher readmission rates?

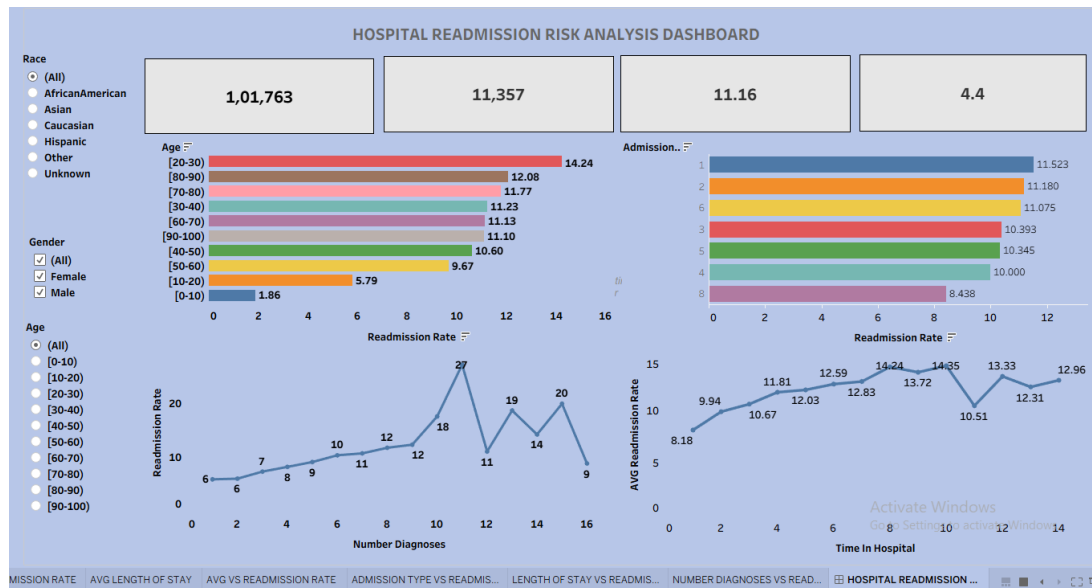
	number_diagnoses	total_patients	readmission_rate
1	1	219	5.94
2	2	1023	6.06
3	3	2835	7.37
4	4	5536	8.26
5	5	11392	9.16
6	6	10161	10.41
7	7	10393	10.77
8	8	10616	11.81
9	9	49473	12.38
10	10	17	17.65

Tools and Technologies

- Python (Pandas, NumPy) for data cleaning and feature engineering
- SQL Server for data analysis and querying
- Tableau for dashboard development and visualization

Dashboard Overview

A single-page interactive dashboard was developed to monitor key readmission indicators and patient risk factors.



Key Performance Indicators (KPIs)

- Total Patients
- 30-Day Readmissions
- Readmission Rate
- Average Length of Stay

Visualizations

- Age Group vs Readmission Rate
- Admission Type vs Readmission Rate
- Length of Stay vs Readmission Rate
- Number of Diagnoses vs Readmission Rate

Key Findings

- Older patients demonstrated higher readmission rates.
- Longer hospital stays were associated with an increased likelihood of readmission.
- Patients with higher prior healthcare utilization showed greater readmission risk.
- Readmission rates increased with the number of diagnosed conditions.
- Certain admission and discharge categories exhibited significantly higher readmission rates.

Conclusion

This analysis identified several important drivers of hospital readmissions, including age, length of stay, prior healthcare utilization, and clinical complexity. The insights generated from this project can help healthcare organizations strengthen discharge planning, improve care coordination, and focus interventions on high-risk patient groups to reduce avoidable readmissions and improve patient outcomes.